1. $n = \frac{2-0}{0.25} = 8$

let $x_0 = 0, x_1 = 0.25, \cdots, x_7 = 1.75, x_8 = 2$

a. $\int_0^2 x^2 \ln(x^2+1)\,dx$

$\approx \frac{0.25}{2}\left[0 + 2\left(\frac{\ln 17}{16} - \frac{\ln 2}{4} + \frac{\ln 5}{4} - \frac{\ln 2}{2}\right.\right.$

$+ \frac{9}{8}\ln 5 - \frac{9}{4}\ln 2 + \ln 2 + \frac{25}{16}\ln 41 - \frac{25}{4}\ln 2$

$\left.+ \frac{9}{4}\ln 13 - \frac{9}{2}\ln 2 + \frac{49}{16}\ln 65 - \frac{49}{4}\ln 2\right)$

$\left.+ 4\ln 5\right]$

≈ 3.1595

b. $\int_0^2 x^2 \ln(x^2+1)\,dx$

$\approx \frac{0.25}{3}\left[0 + 4\left(\frac{\ln 17}{16} - \frac{\ln 2}{4} + \frac{9}{8}\ln 5 - \frac{9}{4}\ln 2\right.\right.$

$\left.+ \frac{25}{16}\ln 41 - \frac{25}{4}\ln 2 + \frac{49}{16}\ln 65 - \frac{49}{4}\ln 2\right)$

$+ 2\left(\frac{\ln 5}{4} - \frac{\ln 2}{2} + \ln 2 + \frac{9}{4}\ln 13 - \frac{9}{2}\ln 2\right)$

$\left.+ 4\ln 5\right]$

≈ 3.1093

c. $\int_0^2 x^2 \ln(x^2+1)\,dx$

$\approx 0.25 \sum_{j=1}^{8} \left\{ (0.25j - \frac{0.25}{2})^2 \ln\left[(0.25j - \frac{0.25}{2})^2 \right] + 1 \right\}$

≈ 3.0842

2.

(a) $\int_0^1 x^2 e^{-x}\,dx$

$= [-x^2 e^{-x}]_0^1 - \int_0^1 (-2x e^{-x})\,dx$

$= -e^{-1} - \{[2x e^{-x}]_0^1 - \int_0^1 2e^{-x}\,dx\}$

$= -e^{-1} - 2e^{-1} + [-2e^{-x}]_0^1$

$= -\frac{5}{e} + 2$

(b) $\int_0^1 x^2 e^{-x}\,dx \approx \frac{1-0}{1}[0 + e^{-1}] = \frac{1}{2e}$

$\approx$

(c) $\frac{d^2}{dx^2}(x^2 e^{-x}) = (2 - 4x + x^2)\, e^{-x}$

$-\frac{-4}{4} = 1 > 0$

$-\frac{1}{12e} < \text{error} < \frac{1}{6} \Rightarrow |\text{error}| < \frac{1}{6}$

bound: $[\frac{5}{12e}, \frac{6+2e}{12e}]$

Absolute error: $|-\frac{5}{e} + 2 - \frac{1}{2e}| = \frac{-11+4e}{2e} < \frac{1}{6}$

(d) $\int_0^1 x^2 e^{-x} dx$

$\approx$

3.

$$\begin{cases} \int_{-1}^{1} 1 \, dx = 2 = a+b \\ \int_{-1}^{1} x \, dx = 0 = -a+b+c+d \\ \int_{-1}^{1} x^2 \, dx = \frac{2}{3} = a+b-2c+2d \\ \int_{-1}^{1} x^3 \, dx = 0 = -a+b+3c+3d \end{cases}$$

$$\Rightarrow \begin{cases} a = 1 \\ b = 1 \\ c = \frac{1}{3} \\ d = -\frac{1}{3} \end{cases}$$

4. $\int_1^{1.5} x^2 \ln x \, dx = \left[\frac{x^3}{3}\ln x\right]_1^{1.5} - \int_1^{1.5} \frac{x^2}{3}\, dx$

$= \frac{9}{8}\ln\left(\frac{3}{2}\right) - \frac{3}{8} + \frac{1}{9}$

$n = 2$:

$\int_1^{1.5} x^2 \ln x \, dx$

$\frac{9}{8}\ln\left(\frac{3}{2}\right) -$

$\int x^2 \ln x + \int \frac{x^2}{3}\, dx$

$= \left[\frac{x^3}{3}\ln x\right]$

$\left[\frac{x^3}{9}\right]_1^{1.5}$

$\frac{3}{8} - \frac{1}{9}$

```python
import math
import matplotlib.pyplot as plt
import numpy as np

global zero

def sign(num):
    if num>0:
        return 1.0
    elif num<0:
        return -1.0
    else:
        print("cannot identify the sign of 0")

def P0(x):
    return np.ones_like(x)

def P1(x):
    return x

def P2(x):
    return (x**2) - 1/3

def P3(x):
    return (x**3) - (3/5) * x

def P4(x):
    return x**4 - (6/7)*(x**2) + 3/35

def Müller(k=2, p_last_1_k = 1, p_last_2_k = -0.9, p_last_3_k = -1):
    global zero
    if k == 2:
        print(0, p_last_3_k)
        print(1, p_last_2_k)

    print(k, p_last_1_k)
    k += 1

    a = (((P4(p_last_1_k) - P4(p_last_2_k)) / (p_last_1_k - p_last_2_k)) - ((P4(p_last_2_k) - P4(p_last_3_k)) / (p_last_2_k - p_last_3_k))) / (p_last_1_k - p_last_3_k)
    b = ((P4(p_last_1_k) - P4(p_last_2_k)) / (p_last_1_k - p_last_2_k)) + ((p_last_1_k - p_last_2_k) * a)
    c = P4(p_last_1_k)

    p_k = p_last_1_k + (-2 * c) / (b + sign(b) * (math.sqrt(b ** 2 - 4 * a * c)))
```

```python
    if P4(p_k) == 0 or abs(p_last_1_k - p_k) < 0.000001:
        print(k, p_k)
        zero = p_k
        return p_k
    else:
        Müller(k, p_k, p_last_1_k, p_last_2_k)

def drawFuncDiagram():
    x = np.linspace(-1, 1, 400)

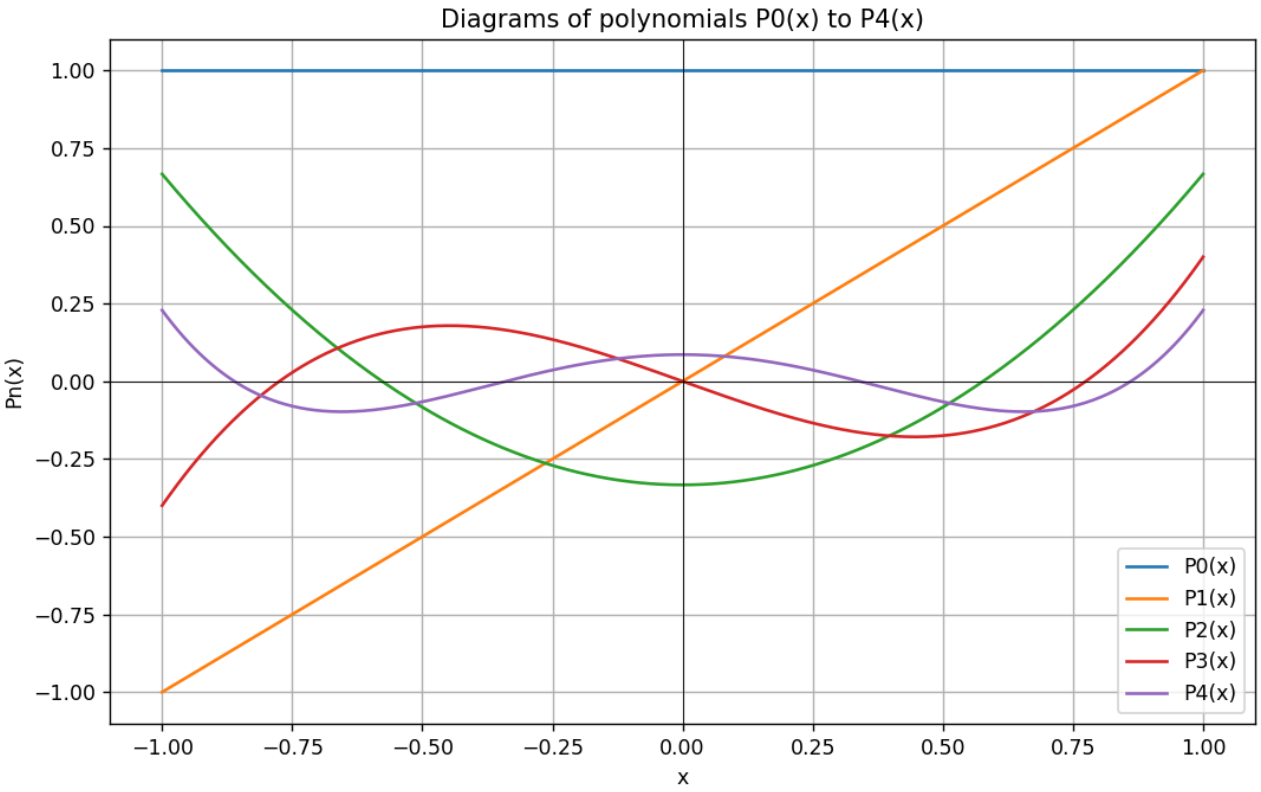
    P0_values = P0(x)
    P1_values = P1(x)
    P2_values = P2(x)
    P3_values = P3(x)
    P4_values = P4(x)

    plt.figure(figsize=(10, 6))
    plt.plot(x, P0_values, label='P0(x)')
    plt.plot(x, P1_values, label='P1(x)')
    plt.plot(x, P2_values, label='P2(x)')
    plt.plot(x, P3_values, label='P3(x)')
    plt.plot(x, P4_values, label='P4(x)')

    plt.title('Diagrams of polynomials P0(x) to P4(x)')
    plt.xlabel('x')
    plt.ylabel('Pn(x)')
    plt.axhline(0, color='black', linewidth=0.5)
    plt.axvline(0, color='black', linewidth=0.5)
    plt.grid(True)
    plt.legend()
    plt.show()

Müller(2,1,-0.9,-1)
zero1 = zero
Müller(2,1,zero1+0.1,zero1+0.01)
zero2 = zero
Müller(2,1,zero2+0.3,zero2+0.1)
zero3 = zero
Müller(2,1,zero3+0.3,zero3+0.1)
zero4 = zero
drawFuncDiagram()
```

```
0 -1
1 -0.9
2 1
3 0.871848576319297
4 0.8642177390444027
5 0.861120916262448
6 0.8611363132243962
7 0.8611363115940528
0 0.8711363115940528
1 0.9611363115940528
2 1
3 0.8606314324259421
4 0.8611614412521825
5 0.8611363174480201
6 0.8611363115940527
0 0.9611363115940527
1 1.1611363115940527
2 1
3 0.8325078041113056
4 0.8657863875067776
5 0.8611966435647094
6 0.8611363370665835
7 0.8611363115940298
0 0.9611363115940298
1 1.1611363115940299
2 1
3 0.8325078041113221
4 0.8657863875067748
5 0.8611966435647094
6 0.8611363370665835
7 0.8611363115940298
```


Diagrams of polynomials P0(x) to P4(x)
Pn(x)
x
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
P0(x)
P1(x)
P2(x)
P3(x)
P4(x)